

Description of Additional Supplementary Files

Supplementary Data 1. Optimized crystal structure of hcp-Li₂ZrCl₆.

Supplementary Data 2. Optimized crystal structure of Li_{2.167}ZrCl_{5.833}O_{0.167}.

Supplementary Data 3. Optimized crystal structure of Li_{2.333}ZrCl_{5.667}O_{0.333}.

Supplementary Data 4. Optimized crystal structure of Li_{2.5}ZrCl_{5.5}O_{0.5}.

Supplementary Data 5. Optimized crystal structure of Li_{2.5}ZrCl_{5.5}O_{0.5}.

Supplementary Data 6. Optimized crystal structure of Li_{2.667}ZrCl_{5.333}O_{0.667}.

Supplementary Data 7. Optimized crystal structure of Li_{2.833}ZrCl_{5.167}O_{0.833}.

Supplementary Data 8. Optimized crystal structure of ccp-Li₂ZrCl₆.

Supplementary Data 9. Optimized crystal structure of Li_{2.125}ZrCl_{5.875}S_{0.125}.

Supplementary Data 10. Optimized crystal structure of Li_{2.25}ZrCl_{5.75}S_{0.25}.

Supplementary Data 11. Optimized crystal structure of Li_{2.375}ZrCl_{5.625}S_{0.375}.

Supplementary Data 12. Optimized crystal structure of Li_{2.5}ZrCl_{5.5}S_{0.5}.

Supplementary Data 13. Optimized crystal structure of Li_{2.625}ZrCl_{5.375}S_{0.625}.

Supplementary Data 14. Optimized crystal structure of Li_{2.75}ZrCl_{5.25}S_{0.75}.

Supplementary Data 15. Optimized crystal structure of Li_{2.875}ZrCl_{5.125}S_{0.875}.

Supplementary Data 16. Initial atomic configuration of hcp-Li₂ZrCl₆ from AIMD simulation at 600 K.

Supplementary Data 17. Final atomic configuration of hcp-Li₂ZrCl₆ from AIMD simulation at 600 K after 300ps.

Supplementary Data 18. Initial atomic configuration of Li_{2.5}ZrCl_{5.5}O_{0.5} from AIMD simulation at 600 K.

Supplementary Data 19. Final atomic configuration of Li_{2.5}ZrCl_{5.5}O_{0.5} from AIMD simulation at 600 K after 300ps.

Supplementary Data 20. Initial atomic configuration of ccp-Li₂ZrCl₆ from AIMD simulation at 600 K.

Supplementary Data 21. Final atomic configuration of ccp-Li₂ZrCl₆ from AIMD simulation at 600 K after 300ps.

Supplementary Data 22. Initial atomic configuration of Li_{2.25}ZrCl_{5.75}S_{0.25} from AIMD simulation at 600 K.

Supplementary Data 23. Final atomic configuration of Li_{2.25}ZrCl_{5.75}S_{0.25} from AIMD simulation at 600 K after 300ps.